

The Dataverse Network: An Infrastructure for Data Sharing

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(8/14/08 talk at “UseR! 2008”, Technische Universität, Dortmund, Germany)

- Gary King, **An Introduction to the Dataverse Network as an Infrastructure for Data Sharing**, *Sociological Methods and Research*, 32, 2 (November, 2007): 173–199.
- Micah Altman and Gary King. **A Proposed Standard for the Scholarly Citation of Quantitative Data**, *D-Lib Magazine*, 13, 3/4 (March/April, 2007).
- Kosuke Imai; Gary King; and Olivia Lau. **Toward A Common Framework for Statistical Analysis and Development**, *Journal of Computational and Graphical Statistics*, forthcoming. (Zelig)
- More information: <http://TheData.org>

Infrastructure for Quantitative Data

- Accessibility:
 - Most large data sets: in public archives
 - Most data in published articles: not accessible, results not replicable without the original author
- Problems even with professional archives:
 - Data in different archives have different identifiers
 - One major archive renumbered all its acquisitions
 - Changes to data are made; identifiers are reused or deaccessioned; old data are lost
- Data sets are not like books
 - Static data files (even if on the web): unreadable after a few years
 - When storage methods change: some data sets are lost; others have altered content!
- Connection to analysis software (like R)
 - uncertain, time consuming, annoying, error prone

What About a Centralized Data Access Solution?

- Highly desirable when feasible
- Works great in astronomy, etc., when data formats are universal, goals are common, and agreements are in place
- **Impossible** when data are heterogeneous in format, origin, size, effort needed to collect or analyze, IRB access rules, etc.
- Why don't researchers put data in public archives?
 - The Archive gets the credit
 - Upon questioning: they want credit, control, and visibility
 - (So why don't they worry about print publishers getting all the credit? Lack of data citations!)
- We propose: **technological solutions to these political problems**

Requirements for Effective Data Sharing Infrastructure

- **Recognition**, for authors, journals, etc. in (1) citations to data, (2) citations to associated articles, and (3) visibility on the web.
- **Public Distribution**, without permission from the author
- **Authorization**: fulfill requirements the author originally met
- **Validation**: check that data exists, without authorization
- **Persistence** Decades from now. . . .
- **Verification**: data remains unchanged, even if converted from SPSS to Stata to R, from a PC to a Mac to Linux, and from 8 inch magnetic tape to 5.25 inch floppies to a DVD.
- **Ease of Use** Neither editors nor authors employ professional archivists
- **Legal Protection**:
 - Journals have liability protection for print; none for data
 - *In the U.S., if you put data on the web without IRB approval, you are violating federal regulations*
 - (IRB approval must be for data distribution, not merely for the study)
 - Solution must not require lawyers (we've automated the IRB)

Rules for Citing Printed Matter

Kim, Jae-On, Norman Nie, and Sidney Verba. 1977. "A Note on Factor Analyzing Dichotomous Variables: The Case of Political Participation," Political Methodology, Vol. 4: No. 2 (Spring): Pp. 39–62.

First author (last name first) Second author Third author Year Article title Journal (no longer exists) Volume number Issue number Season Pages
Special formatting codes Special indentation Citations: rule-based, precise, redundant Print Citations Work: authors don't think publishers get all the credit; cited articles can be found; copyeditors don't need to see the original to know it exists; the link from citation to print persists

A New Citation Standard for Numeric Data

Sidney Verba, 1998, "Political Participation Data", [hdl:1902.4/00754](https://hdl.handle.net/1902.4/00754),
UNF:3:6:ZNQRI14053UZq389x0Bffg?== Annals of Applied Statistics
[Distributor]; NORC [Producer].

- 1 Author
- 2 Year
- 3 Title
- 4 Unique Global Identifier: will work after URLs stop working
- 5 Linked to a Bridge Service (presently a URL:
<http://id.thedata.org/hdl%3A1902.4%2F00754>)
- 6 Universal Numeric Fingerprint (UNF)
- 7 Standard rules for adding citation elements

Data to Universal Numeric Fingerprints

$$\begin{pmatrix} 1 & 4 & 4 & 21 & \dots & 121 \\ 1 & 2 & 2 & 91 & \dots & 212 \\ 1 & 9 & 2 & 72 & \dots & 104 \\ 0 & 2 & 2 & 2 & \dots & 321 \\ 1 & 6 & 2 & 12 & \dots & 204 \\ 1 & 9 & 4 & 52 & \dots & 311 \\ 0 & 3 & 2 & 23 & \dots & 92 \\ 0 & 2 & 5 & 91 & \dots & 212 \\ 0 & 5 & 8 & 91 & \dots & 91 \\ 1 & 9 & 1 & 72 & \dots & 104 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 2 & 2 & 91 & \dots & 212 \end{pmatrix} \Rightarrow \text{ZNQRI14053UZq389x0Bffg?}==$$

Advantages of UNFs

- UNF is **calculated from the content** not the file: Its the **Same UNF** regardless of changes in computer hardware, storage medium, operating system, statistical software, database, or spreadsheet software.
- Cryptographic technology: **any** change in data content changes the UNF. (cannot tinker after the fact!)
- Noninvertible properties
 - UNFs convey no information about data content
 - OK to distribute for highly sensitive, confidential, or proprietary data
 - Copyeditor can validate data's existence even without authorization
- The citation refers to **one specific data set** that can't **ever** be altered, even if journal doesn't keep a copy
- Future researchers can quickly check that they have the same data as used by the author: merely recalculate the UNF

Web 2.0 Terminology

- **Software**: find CD, install locally, hit next, hit next, hit next. . .
- **Web application software**: no installation; load web browser and run (Dataverse Network Software)
- **Host**: The computers where the web application software runs (universities, archives, libraries)
- **Virtual host**: Where the web application software *seems* to run, but does not (web sites of: authors, journals, granting agencies, research centers, universities, scholarly organizations, etc.)

<http://www.peterson.com>

<http://dvn.iq.harvard.edu/peterson>

Bob Peterson
Professor of Government

Blog: Why the OSA is wrong this time on open-source projects
January 30, 2008, 9:37 AM PST

"More open-source projects have originated in Europe than anywhere else in the world," said Bertrand Diant, chief executive officer of data-integration specialist Talent, a founding member of the Open Solutions Alliance. Founded a year ago, the OSA has had a U.S. focus until now.

The European chapter of the OSA will be formally incorporated in the next 90 days, and will then look for interoperability work required by European users.

"When the OSA got started, we saw our mission as a global one, but the critical mass of activity so far has been in the U.S.," said Dominic Sartorio, OSA president and senior director of product management at service company SpikaSource. "We were approached two months ago by a group of companies in Europe thinking of forming a group like ours but focused on Europe. This led us to think that while we are trying to achieve results globally, there are some parts of the world where we are doing less well."

Sartorio sees 2008 as turning point for open source. In late 2007, he promised that OSA would "out-Microsoft Microsoft" in response to a user survey that revealed a demand for interoperability among

Current Project: Political Economy in Ghana

In the last 20 years the economics profession has been revolutionized by political economy. For example, until the mid 1980s in traditional public finance or macroeconomics, the role of economists was to design socially optimal policies faced with various types of trade-offs. The fact that actual observed policies deviated, often radically, from those proposed by economists was not systematically explained other than by the implicit notion that policymakers did not understand what the best policy was. In addition, many explanations for underdevelopment went in terms of market failures, such market failures were taken as exogenous and their implications derived.

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STUDIES

Search Cataloging Information for

10 Peterson Datasets

- ☐ 10 Million International Dyadic Events by Gary King, Will Lowe
- ☐ Causes of Death Data by Frederick Groves, Gary King
- ☐ Elections to the United States House of Representatives, 1898-1982 by Gary King
- ☐ Application data for: A Solution to the Ecological Inference Problem: Reconstructing Individual Behavior from Aggregate Data by Gary King
- ☐ Application data for: A Statistical Model of Multi-Party Electoral Data by Jonathan Katz, Gary King
- ☐ Application data for: A Unified Model of Cabinet Dissolution in Parliamentary Democracies by Gary King, James E. Alt, Penny Burns, and Michael Laver
- ☐ Application data for: Constituency Service and Incumbency Advantage by Gary King
- ☐ Application data for: Death by Survey: Estimating Adult Mortality without Selection Bias from Sibling Survival Data by Emmanuelle Gakidou, Gary King
- ☐ Application data for: Detecting Model Dependence in Statistical Inference: A Response by Gary King, Langche Zeng
- ☐ Application data for: Enhancing Democracy Through Legislative Redistricting by Andrew Gelman, Gary King
- ☐ Application data for: Estimating the Probability of Events That Have Never Occurred: When is Your Vote Decisive? by Andrew Gelman, Gary King, and Justin Boykoff
- ☐ Application data for: Explaining Rare Events in International Relations by Gary King, Langche Zeng
- ☐ Application data for: Improving Forecasts of State Failure by Gary King, Langche Zeng
- ☐ Application data for: Improving Quantitative Studies of International Conflict: A Conjecture by Nathaniel Beck, Gary King, and Langche Zeng
- ☐ Application data for: Making the Most of Statistical Analyses: Improving Interpretation and Presentation by Gary King, Michael Tutz, and Jason Wittenberg
- ☐ Application data for: Mapping as Nonparametric Preprocessing for Reducing Model Dependence in Parametric Causal Inference by Daniel E. Ho, Kosuke Imai, Gary King, Elizabeth A. Stuart
- ☐ Application data for: Measuring the Consequences of Delegate Selection Rules in Presidential Nominations by Stephen Anagnostis, Gary King
- ☐ Application data for: On Party Platforms, Mandates, and Government Spending by Gary King, Michael Laver

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REPLICATION DATA FOR: IMPROVING FORECASTS OF STATE FAILURE

Cataloging Information Documentation, Data and Analysis

Citation Information

How to Cite

Study Global Id

Authors

Production Date

Distributor

Distributor Contact

Deposit Date

Replication For

Provenance

Abstract and Scope

We offer the first independent scholarly evaluation of the claims, forecasts, and causal inferences of the State Failure Task Force and their efforts to forecast when states will fail. State failure refers to the collapse of the authority of the central government to impose order, as in civil wars, revolutionary wars,

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REPLICATION DATA FOR: IMPROVING FORECASTS OF STATE FAILURE

Cataloging Information **Documentation, Data and Analysis**

Download all files in a single archive file (files that you cannot access will not be downloaded)

File Name	Description	Cases/ Variables	Type	Controls
1. Documentation				
improvingforecasts.pdf	Article related to this study: Improving Forecasts of State Failure		application/pdf	
2. Replication Data				
rv00.apic	Sample spec file for committee n's			
rv01.dat	For use with committee n's, ASCII case file			
rv01.tab	For use with committee n's	7190 11	Tab delimited	
replication.txt	Additional document describing the format of the replication files			
3. Recoded Data				
names.dta	Country names, in original Stata format			
names.tab	Country Names Data	8580 3	Tab delimited	
sort1rv01a.dat	Reordered Full Data in Original Format, comma delimited text file			
4. Original Data				
rv01rv01a.pdf	Documentation for Original		application/pdf	

http://yourwebsite.com

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REPLICATION DATA FOR: CONSTITUENCY SERVICE AND INCUMBENCY ADVANTAGE
DATA FILE: CONSTITUENCYSERVICE.TAD

Back to Study

Download Subset Subset and Recode Descriptive Statistics Advanced Statistical Analysis

Selected Variables

To recode (subset) a variable into a different one, first, select a variable from the selected-variables box, click the arrow button below, and then the name and label of the variable you have chosen appear in the new variable name and label boxes for convenience. You must replace the old variable name with a unique variable name that is not used in the data file; the new variable label is optional and you can leave it blank.

New Variable Name
New Variable Label

Apply Recodes

Select variables from table below (selected variables will be displayed above) You do not have access to the subsetting and analysis functionality for this restricted data file. You can only view the variables and their summary statistics.

Show 20 Variables

Variable Type	Variable Name/Variable Label	Quick Summary
☐ Continuous	INCAD Incumbency advantage	100
☐ Continuous	BUDG budget figures	100
☐ Continuous	SAL salary	100
☐ Discrete	COQ state level dummy variables	100

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Dataverse

The screenshot shows a web browser window displaying the Dataverse Network website. The header features a portrait of Bob Peterson, Professor of Government, and the text "Bob Peterson Professor of Government". Below this, there is a navigation bar with links for "Search Dataverse", "User Guides", "Site Map", "Contact Us", and "Log In". The main content area displays the title "REPLICATION DATA FOR CONSTITUENCY SERVICE AND INCUMBENCY ADVANTAGE" and the data file "CONSTITUENCYSERVICE.TAB". A sidebar on the left contains links for "Bio/CV", "Writings", "Current Classes", "Contact", and "Dataverse". The main content area includes a "Download Subset" button and a "Choose a Statistical Model" dropdown menu. A list of statistical models is displayed, including "Negative Binomial Regression for Event Count Dependent Variables", "Poisson Regression for Event Count Dependent Variables", "Generalized Additive Model for Event Count Dependent Variables", "General Estimating Equation for Poisson Regression", and "Social Network Poisson Regression for Event Count Dependent Variables".

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REPLICATION DATA FOR CONSTITUENCY SERVICE AND INCUMBENCY ADVANTAGE
DATA FILE: CONSTITUENCYSERVICE.TAB

Download Subset Subset and Records Descriptive Statistics Advanced Statistical Analysis

Selected Variables

Choose a Statistical Model

Choose a Statistical Model

Categorical Data Analysis

Cross-Tabulation

Ecological Inference Models

Hierarchical Multinomial-Dirichlet Ecological Inference Model for R x C Tables

Selected Event Count Models

variables are

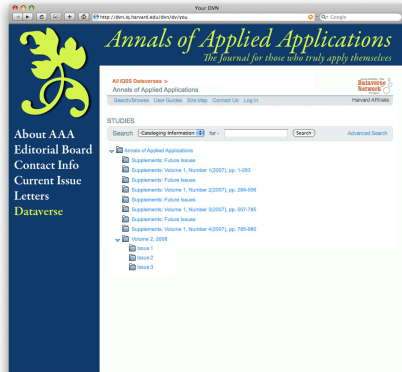
- Negative Binomial Regression for Event Count Dependent Variables
- Poisson Regression for Event Count Dependent Variables
- Generalized Additive Model for Event Count Dependent Variables
- General Estimating Equation for Poisson Regression
- Social Network Poisson Regression for Event Count Dependent Variables

Variable

- Continuous Models for Continuous Bounded Dependent Variables
- Continuous Exponential Regression for Duration Dependent Variables
- Continuous Gamma Regression for Continuous, Positive Dependent Variables
- Discrete General Estimating Equation for Gamma Regression
- Discrete CT state level dummy variables
- Discrete DE state level dummy variables
- Discrete IA state level dummy variables
- Discrete MI state level dummy variables
- Discrete MO state level dummy variables
- Continuous SAL salary
- Discrete CO state level dummy variables



Your web site



Your dataverse branded as your web site but served by the Dataverse Network, therefore re-quiring no local installation and providing an enormous array of services

The Dataverse Network Project Homepage (<http://TheData.org>)

Dataverse Networks may harvest metadata from each other (dashed lines)

The IQSS Dataverse Network at Harvard University

[Other] Dataverse Network[s]

SMR Dataverse

Gary King's Dataverse

PoliSci 101 Dataverse

U.S. Census Dataverse

APSA Legislative Section
Dataverse

NSF Dataverse

Weatherhead Center
Dataverse

MacArthur Network on
Inequality Dataverse

Dept of Psychology
Dataverse

Dataverse owners may list, or allow searching across, other dataverses or the data sets in them

Each Dataverse Network may serve numerous individual dataverses (arrows)

Most users come directly to a dataverse, which is a self-contained archive (with all services provided by a Dataverse Network)

Software is available at the project homepage and only needs to be installed to establish a Dataverse Network. Dataverses are virtual hosts.

Your Dataverse

- **Full service virtual archive**, with numerous data services (citation, metadata, archiving, subsetting, conversion, translation, analysis, ...)
- **List of your data, or your view of the universe of data**
- **Branded as yours**: with the look and feel of your site
- **Easy to setup**: give DVN your style, and include a link to your new dataverse
- **Easy to manage**: no software or hardware installation, backups, worry about archiving standards, or data format transations; still exists if you move; easy to rebrand
- **High acceptability**: experiments indicate $> 90\%$ uptake for authors
- **Reuse**: same data may appear on different dataverses
- **Results**: **Articles with data available have twice the impact factor!**
(with dataverse, it should be more)

Dataverse Uses

- Authors, for their data or their view of the universe of data
- Journals, for replication data archives
- Future Researchers: browse or search for a dataverse or dataset; forward citation search; verification via UNFs; subsetting; read metadata, abstract, & documentation; check for new versions; translate format; statistical analyses; download
- Teachers, a list or for in depth analysis
- Sections of scholarly organizations, to organize existing data
- Granting agencies
- Research centers
- Major Research Projects
- Academic departments, universities, data centers, libraries
- Data archives

The Universe of Data meets the Universe of Methods

- **R Project for Statistical Computing**

- nearly 1000 packages; most new methods appear in R first
- Highly diverse examples, syntax, documentation, and quality
- Can be difficult for us; harder for applied researchers

- **Zelig: Everyone's Statistical Software**

- An ontology we developed of almost all statistical methods
- Users incorporate original packages a simple model description language (and R bridge functions)
- Result: Unified Syntax, **the same 3 commands to use any method**
- Easy for applied data analysts who use R

- **R + Zelig + Dataverse Network**

- Write Zelig bridge function $\rightsquigarrow$ your method appears in the DVN GUI
- **Greatly reduced time from methods development to widespread use**
- Easy for applied researchers who don't use R
- (GUI time not wasted: save R code for replication or further analysis)

How to participate

- To increase citations to your data (& web visibility), choose:
 - Sign up for a free [dataverse](#) for your web site (no installations, branded as yours, citations for all your data)
 - or install [DVN software](#) & you can also give out dataverses
- To increase use of your R package through Zelig and the DVN GUI:
 - Write a simple Zelig bridge function
- To join us:
 - DVN and Zelig are open source projects; contributions welcome!
- For more information:

<http://TheData.org>

Technology used in DVN Software

- Language: **Java Enterprise Edition 5** (with EJB3 and JSF) (team picked for JavaOne; Sun engineers regularly call for advice)
- Application server: **GlassFish** (wrote press release on our project)
- Database: we use **PostgreSQL** (can substitute others)
- Statistical computing: **R** and **Zelig**